Adenine phosphoribosyltransferase deficiency presenting with supposed 'uric acid' stones: pitfalls of diagnosis[1]

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Adenine is not a normal constituent of body fluids (Simmonds 1969) and is usually excreted at levels below the limit of detection by most methods. The identification of adenine in quantity in the urine of a 2½-year-old boy (B Dh) (Simmonds *et al.* 1976*a*) was the first indication that led to the correct identification of crystals and stones passed since birth by this child as 2,8-dihydroxyadenine, and not uric acid. An identical case was reported independently in a 4-year-old boy in France (Cartier & Hamet 1974). Here, too, the stones had originally been identified as uric acid stones. The metabolic basis for the defect was later identified in both cases as being due to an almost total defect of the adenine salvage enzyme, adenine phosphoribosyltransferase (EC 2.4.2.7, APRT) (Debray *et al.* 1976, Van Acker *et al.* 1977).

This paper reports the identification of 2,8-dihydroxyadenine as the principal component in stones taken from the renal pelvis and ureter of a 19-month-old Arab girl (S Rz). The difficulties inherent in the usual methods of stone analysis, which lead to the confusion of these stones with uric acid stones, are pinpointed. Since early diagnosis could be important therapeutically in these children, two separate systems which could assist in the correct identification of the stone material are reported (Westbury & Omenogor 1970).

Clinical details

The patient, S Rz, is the third child of consanguinous parents. She was well until one year of age, when she began to suffer repeated attacks of abdominal pain and fever. After eight months of these attacks, urinary tract infections were diagnosed, and intravenous urography showed a non-functioning left kidney and a non-radio-opaque stone in the pelvis of the right kidney. About this time she passed a stone, reported as brown in colour. In April 1977, at the age of 19 months, she was admitted for investigation.

Cystoscopy and retrograde ureteric catheterization revealed a block at 2 cm above the lower end of the left ureter. This calculus was removed by ureterolithotomy, and the stone in the right renal pelvis was removed at the same operation; both stones were sent for analysis.

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Clinical details of the first patient (B Dh) and his family have already been published (Van Acker *et al.* 1977). Full clinical details of this patient (S Rz) will be published with the family study (Barratt *et al.* 1978).

Methods

Stone analysis was performed first by the thermogravimetric technique (Rose & Woodfine 1976) and later by a quantitative wet chemistry technique (Westbury & Omenogor 1970). Special methods used for the identification of 2,8-dihydroxyadenine in the stones and urine have also been reported previously (Simmonds 1969, Simmonds *et al.* 1976*b*, Van Acker *et al.* 1977).

The technique of isotachophoresis of the purines has been worked out by us and is in preparation for publication. In this technique the urinary components separate according to their mobility, being moved at constant velocity by voltage increment in an apparatus with few moving parts which does not require an expert for operation. The machine used was an LKB Tachophor 2127. The terminating electrolyte was β-alanine, pH 10.6; the leading electrolyte was 2.5 mmol Tris-HCl, pH 7.9 in 0.3% hydroxypropylmethylcellulose. Five microlitre samples of urine, diluted where necessary, were applied directly. The time for total separation varied from 40–45 minutes. Ultraviolet absorbance (recorded as % transmission) was measured at 254 nm and at a chart speed of 8 cm/min.

Results

In appearance the stone was round and rough; it was friable and pale grey on crushing. The original examination by thermogravimetric analysis gave a tracing virtually indistinguishable from that of uric acid (Figure 1) (as did the stone of our previous case, B Dh). The identity of the stone of S Rz as 2,8-dihydroxyadenine and not uric acid was first established by u.v. spectrophotometry. (It is now routine practice to confirm the composition of all uric acid stones from children by this means.) Infrared spectrophotometry (Figure 2) established conclusively that the principal stone component was 2,8-dihydroxyadenine and not uric acid. The technique of quantitative wet chemistry showed a positive colorimetric test for 'uric acid' (accounting for 88% of the stone material) principally in the acid fraction. This is the reverse of the situation with uric acid stones where the principal component appears in the alkaline

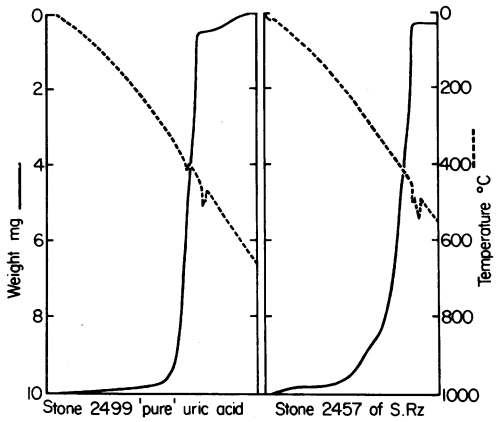


Figure 1. Thermogravimetric tracings from a pure uric acid stone (left) and the stone of S Rz (right) to show their similarities. These tracings show progressive weight loss on linearly raising the temperature, and in the two stones the main losses are at the same temperature

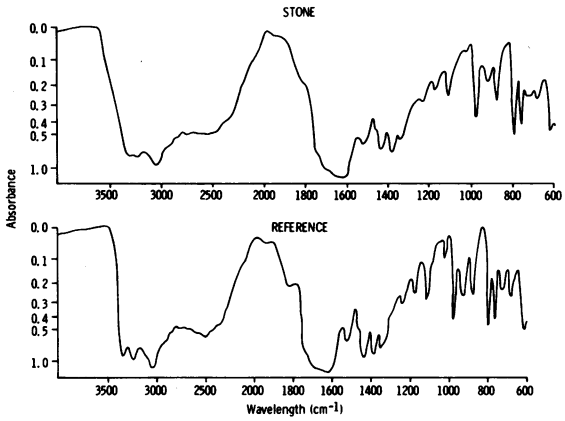


Figure 2. Infrared spectrum of fragments of the stones from S Rz (upper) compared with a reference sample of 2,8-dihydroxyadenine, examined as potassium bromide discs in a Perkin Elmer 377 Diffraction Grating Spectrophotometer in the Wave number range 625–4000 cm^{-1}

fraction. This test may thus prove a useful adjunct to thermogravimetric analysis for the initial separation of these two compounds.

Figure 3A illustrates the scans obtained by isotachophoresis of the urine in the homozygote (S Rz), her heterozygous mother, and the same sample from the mother to which standards had been added before repeating the scan. Figure 3B compares the scans obtained in a control child and the two siblings in a previous family (B Dh and F Dh) (Van Acker *et al.* 1977, Simmonds *et al.* 1977). The scan obtained in the case of the heterozygote is indistinguishable from that of the control; those of the homozygotes are characterized by one marked difference – a large predominant adenine peak. (2,8-dihydroxyadenine and 8-hydroxyadenine are not detectable at 254 nm. The small peak in the adenine position is due principally to impurities in the buffer which migrate in the terminator.) Asymptomatic homozygotes (such as F Dh) may therefore be detected by the adenine excreted.

Erythrocyte enzyme studies in both whole cells and lysates (Dean *et al.* 1978) have confirmed the almost total deficiency for the adenine phosphoribosyltransferase (APRT) enzyme in S Rz, the parents being heterozygotes for the defect.

Discussion

In this case (S Rz) of APRT deficiency, as in both previous cases (Cartier & Hamet 1974, Simmonds *et al.* 1976*a*), the stone was at first incorrectly reported as consisting of uric acid. This was due to the fact that it is impossible to distinguish between uric acid and the real stone component, 2,8-dihydroxyadenine, by the usual methods of stone identification. This could have had unfortunate consequences because treatment with alkali to dissolve uric acid stones will be ineffective with these stones and could even be harmful, as will be discussed later (the solubility of 2,8-dihydroxyadenine does not vary within the physiological pH range (Debray *et al.* 1976)). It is therefore important to consider how such mistakes could be avoided in the future.

First, the appearance and structure of the stone in both cases was quite different from that of uric acid stones. The latter are usually smooth, hard and pale yellow; the stones in APRT deficiency are rough, friable and pale grey on crushing (Simmonds *et al.* 1976*a*).

Second, although thermogravimetric analysis, a quantitative technique of stone analysis, will not separate the two compounds, nor may they be distinguished by the usual colorimetric tests (in which 2,8-dihydroxyadenine reacts mole for mole as uric acid), it is possible to identify them by a system based on wet chemistry. In this system, the chromogenic material appears in

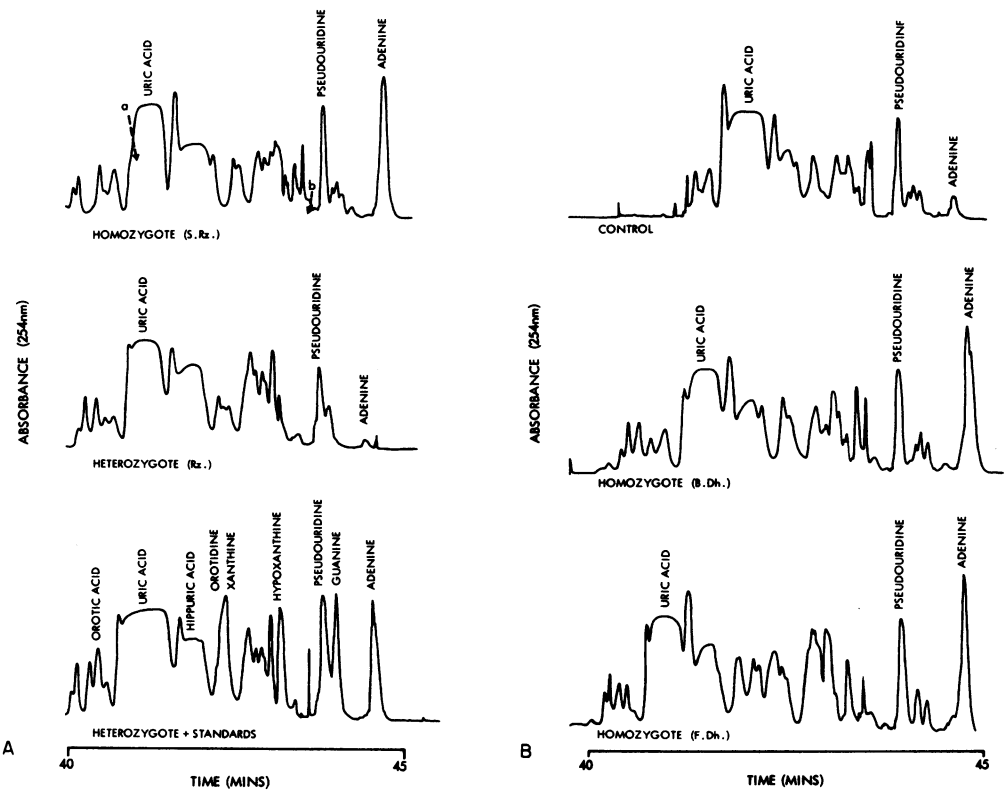


Figure 3. Scans obtained by isotachophoresis of the urine using an LKB 2127 Tachophor. The u.v. absorbance was recorded at 254 nm and at a chart speed of 8 cm/min. A, compares the scans from the homozygote (S Rz) (the dotted arrows indicate the position for 2,8-dihydroxyadenine (a) and 8-hydroxyadenine (b) not detectable at 254 nm) and her heterozygous mother. The lower scan is a repeat scan from the mother to which standards have been added. B, compares the two homozygotes (B Dh and F Dh) from the previous report (Van Acker *et al.* 1977) with a scan from the urine of a control child

the acid rather than the alkaline extract in the case of 2,8-dihydroxyadenine stones, and vice versa with uric acid stones. This could be useful where positive means of final identification, such as infrared and mass spectrometry, are not available.

Third, isotachophoresis, which requires only a few drops of a random urine sample, will indicate the presence or absence of adenine and hence homozygosity for APRT deficiency. This technique could, therefore, be useful in suspected cases of 'uric acid' stones in the paediatric age range. It could be invaluable in the detection of asymptomatic homozygotes such as F Dh, the clinically normal brother of the propositus in our earlier family (Van Acker *et al.* 1977). However, it will not detect heterozygotes for the defect.

The renal complication in these children has previously been shown to be the direct result of the enzyme defect; in the absence of the APRT enzyme no other routes for the disposal of adenine are available in mammalian tissues and the only alternative route is oxidation by xanthine oxidase to the extremely insoluble 2,8-dihydroxyadenine (Van Acker *et al.* 1977, Simmonds *et al.* 1977). The xanthine oxidase inhibitor, allopurinol, should therefore be the appropriate form of therapy. The stress we place upon the importance of early diagnosis is based on the following observations:

(1) The finding of another homozygote for APRT deficiency in such a short space of time suggests the defect could be more prevalent than hitherto suspected. Two separate reports which have screened large sections of the population for APRT deficiency have found the

incidence of heterozygotes to be as high as 0.5–1.0% (Emmerson *et al.* 1977, Fox 1977), which would support the above suggestion.

(2) Alkali administration is traditional therapy for uric acid urolithiasis. As indicated earlier, alteration of urinary pH within the physiological range will not improve the solubility of 2,8-dihydroxyadenine and may even be harmful. For example, allopurinol was ineffective when given previously together with alkali in the case of B Dh, although the child has now been stone-free for more than 18 months on allopurinol alone (Van Acker *et al.* 1977). The propositus in this report may have had access to antacid tablets for several months. For this reason it would appear wise to withhold alkalis in this condition.

It is therefore recommended that when 'uric acid' is found as the main constituent of renal calculi in children, by either qualitative or thermogravimetric analysis, this should be confirmed by appropriate solubility studies. Where possible such stones should be submitted for specialist analysis. Alternatively, or in addition, the identification of adenine in the urine, for instance, by isotachophoresis, will confirm homozygosity for a deficiency of the APRT enzyme.

Summary

Fragments of stones removed from the ureter and renal pelvis of a 19-month-old girl, initially identified as uric acid, were shown to consist almost totally (>80%) of 2,8-dihydroxyadenine, an insoluble oxidation product of adenine. The formation of these stones, as in two previous cases, was due to complete deficiency of the purine salvage enzyme adenine phosphoribosyltransferase (APRT). Suggestions as to how this error in stone identification may be avoided in the future include comparative solubility studies, as well as the use of a simple urinary screening test for the detection of homozygotes for the defect. It is advised that all 'uric acid' stones within the paediatric age range be subjected to specialist analysis.

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